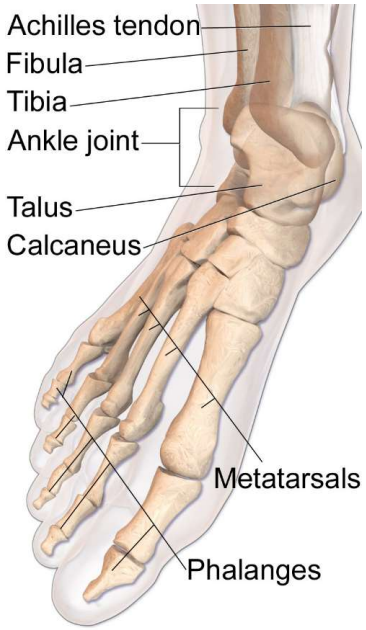
Human foot

- Mechanical marvel
- 26 bones, 33 joints, and more than 100 tendons, muscles, and ligaments
- One of the most complex parts of the human body
- Multi-functional



Lower Leg and Foot

Blausen.com staff (2014). "Medical gallery of Blausen Medical 2014". WikiJournal of Medicine 1 (2). DOI:10.15347/wjm/2014.010. ISSN 2002-4436. - Own work, CC BY 3.0, https://commons.wikimedia.org/w/index.php?curid=29849183

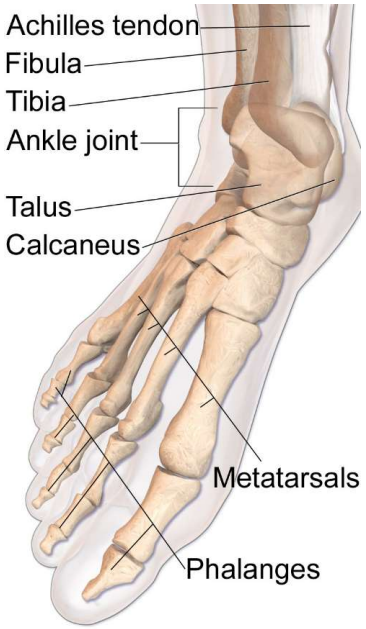
Human foot vs Prosthetic feet

Endoskeletal – Exoskeletal



SACH foot (Otto Bock)

Solid Ankle Cushioned Heel



Lower Leg and Foot

Blausen.com staff (2014). "Medical gallery of Blausen Medical 2014". WikiJournal of Medicine 1 (2). DOI:10.15347/wjm/2014.010. ISSN 2002-4436. - Own work, CC BY 3.0, https://commons.wikimedia.org/w/index.php?curid=29849183



http://newsfeed.time.com/2013/02/23/sum-of-the-parts-prosthetics-through-the-centuries/slide/inside-foot/

Jaipur foot

Prosthetic feet



Pyramid
Pyramid receiver



Pyramid adapter

Keel



Magnum® Foot

Keel



Carbon Copy 2® Foot

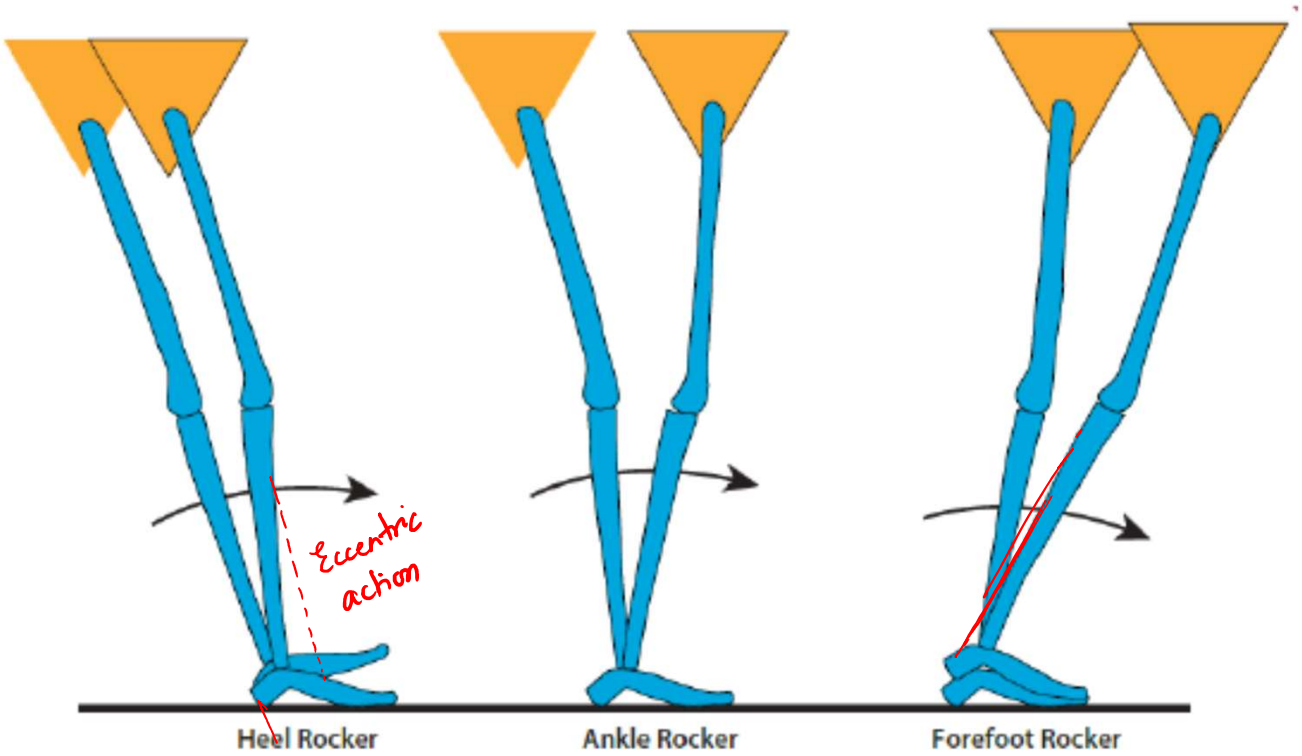
Understanding function

- A prosthesis is a replacement
- Understand function of the limb it replaces
- Understand *what* the function is, not *how* the human body accomplishes it
- Need for compensatory effects
- Aim to restore independence

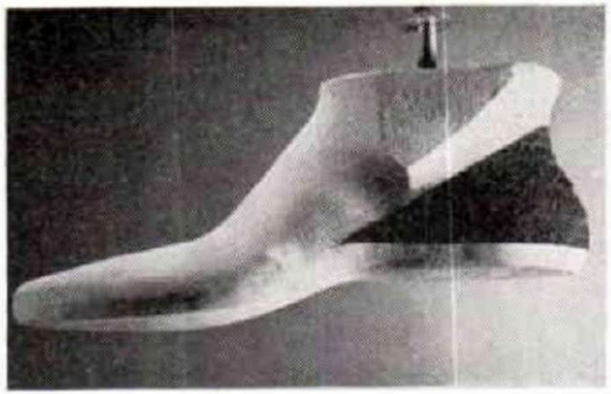
Role of a lower limb prosthesis

- Provide stability in standing and walking
- Shock absorption
- Provide ability to sit comfortably
- Enable forward motion of leg to prepare for the next step
- Assist in push-off
- Control pendular swing
- Adjust to different walking speeds
- Reduce effort needed to walk
- Restore independence
- Cater to other specialized functional needs - swimming, sprinting

Muscle action at foot and ankle during gait



Prosthetic foot at heel strike



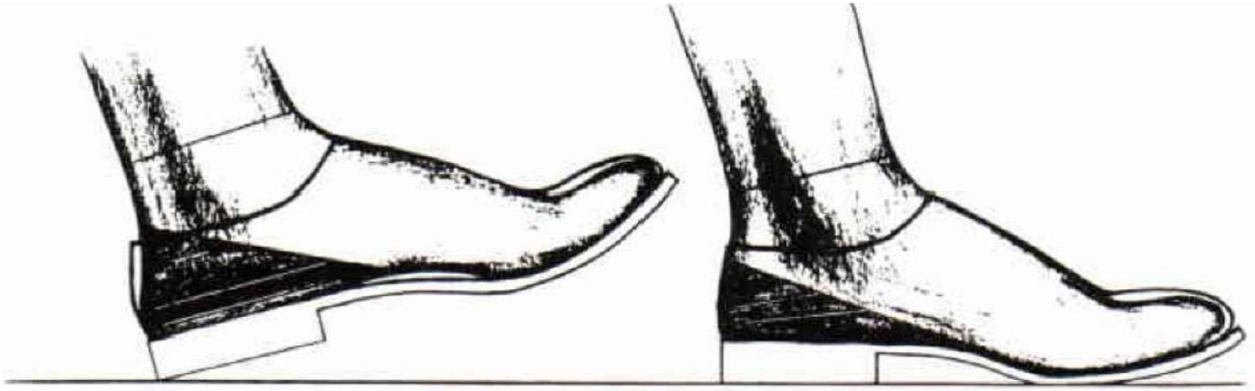
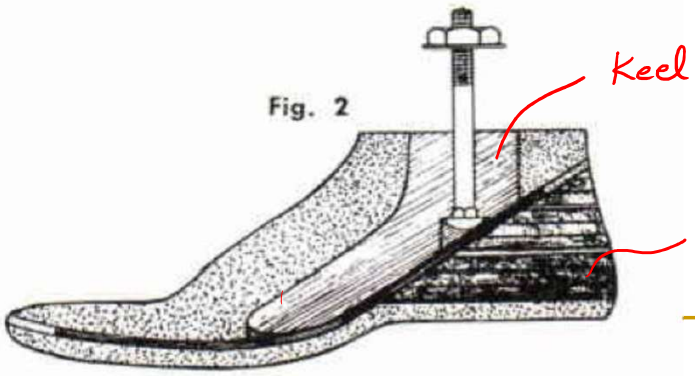


Figure 1—Simulation of plantar flexion by heel cushion compression.

1957.The SACH (Solid-Ankle Cushion-Heel) Foot by ANTHONY STAROS,
ORTHOPEDIC & PROSTHETIC APPLIANCE JOURNAL



Prosthetic foot – energy storage and return

ESAR

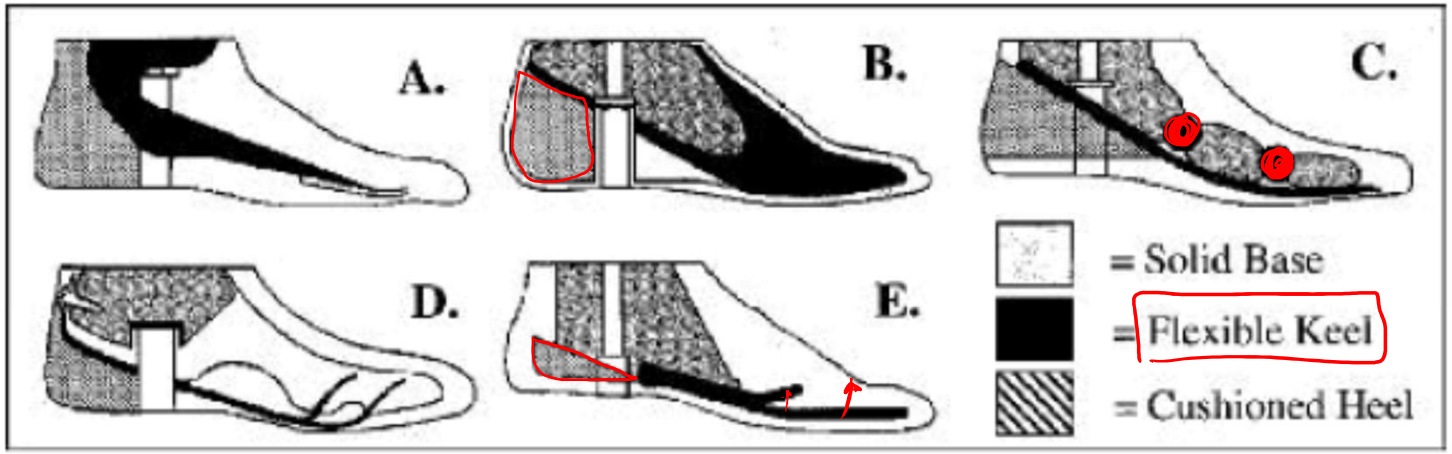


Figure 1.
Crosssections of various energy-storing feet. Each foot is composed of a compressible heel and flexible keel spring. A. Seattle Foot, B. Dynamic Foot, C. STEN Foot, D. SAFE Foot, E. Carbon Copy II Foot.
Image based on the images of Wing DC, Hittenberg DA. Arch Phys Med Rehabil 1989;70(4):330–5.

Transtibial energy-storage-and-return prosthetic devices: A review of energy concepts and a proposed nomenclature
Brian J. Hafner, BS; Joan E. Sanders, PhD; Joseph M. Czerniecki, MD; John Fergason, CPO
Journal of Rehabilitation Research and Development
Vol. 39 No. 1, January/February 2002
Pages 1 - 11

Prosthetic foot – energy storage and return

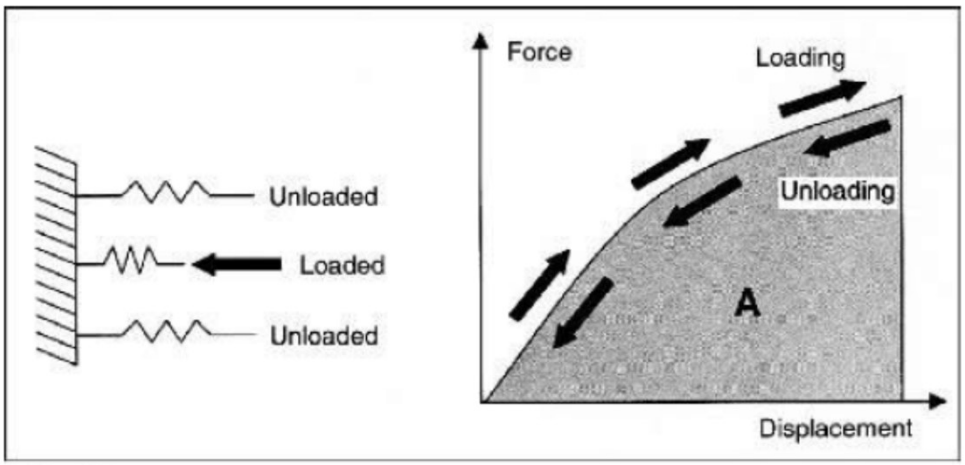


Figure 3.
Potential energy derived from a spring in compression corresponds to area A.

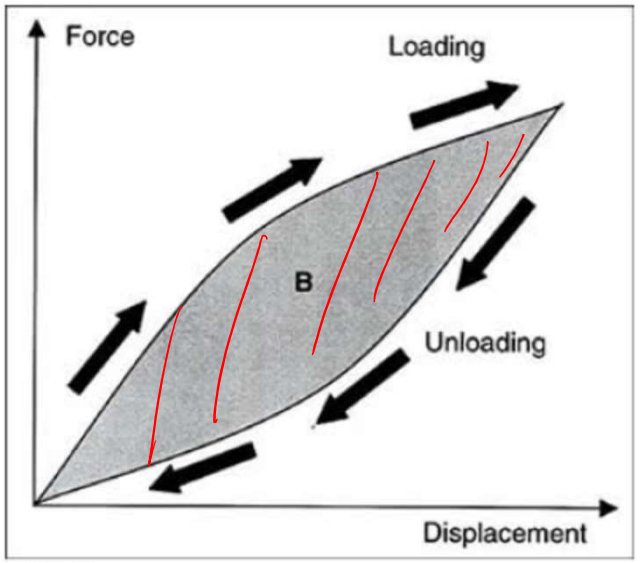


Figure 4.
Simulated load-deformation curve of a viscoelastic material.
B = energy lost as a result of friction.

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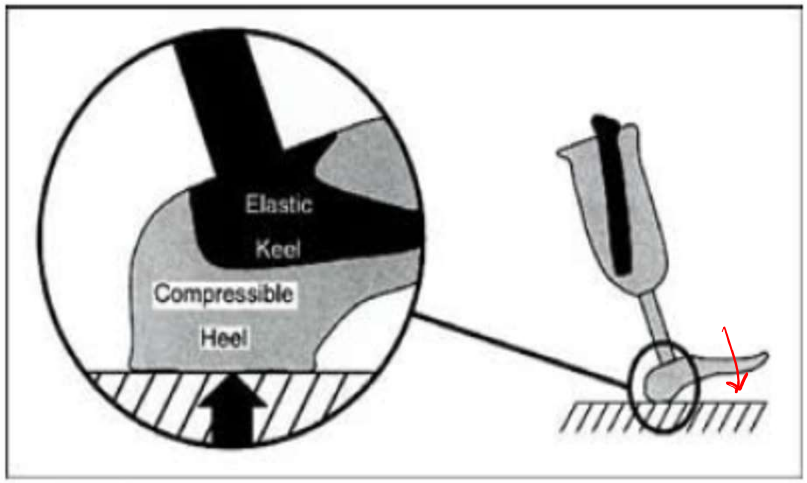


Figure 6.
Heel compression of a prosthesis during loading response.

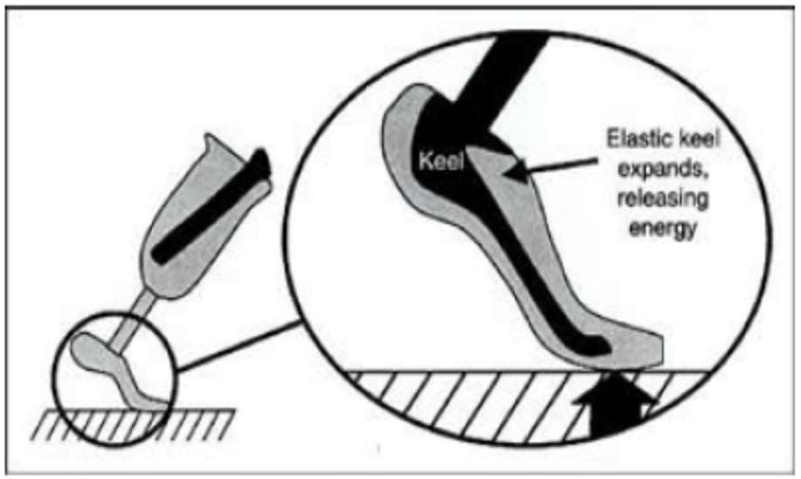


Figure 7.
Keel loading of a prosthesis during terminal stance.

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Prosthetic feet

SACH foot (Otto Bock)





Trailblazer® MA Foot



Pathfinder II Foot





Single Axis foot (WillowWood)



http://newsfeed.time.com/2013/02/23/sum-of-the-parts-prosthetics-through-the-centuries/slide/inside-foot/

Prosthetic feet



Fusion® Foot



Pathfinder II Foot



Trailblazer® MA Foot